

Hydrologic Response Units in OpenHydroQual

The HRU and Urban HRU composite blocks

Abstract

A hydrologic response unit (HRU) in OpenHydroQual is a composite block that bundles the surface, soil and groundwater stores of one piece of land and the connectors between them, so that a sub-catchment is added to a model as a single object with a single set of properties. This document describes the two HRUs: the `Hydrologic_Response_Unit` (“HRU”), a single infiltrating surface over a three-layer soil column and a groundwater cell, and the `Urban_HRU`, which divides the land into pervious and impervious surfaces with their own routing reaches, places the groundwater cell under the whole unit as an unconfined aquifer whose initial state is a water-table elevation, recharges it through the unsaturated conductivity of the bottom soil layer, can draw evapotranspiration from the deeper soil layers through a second source, takes soil parameters layer by layer and carries three calibration multipliers. For each unit the document sets out the members and connectors, the equation of each flux, how the plan area is divided, how the vertical geometry and the initial state follow from a few properties, and how the impervious fraction should be estimated, and it closes with a worked example and a property reference.

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1 Which HRU to use

	Hydrologic_Response_Unit	Urban_HRU
File	hydrologic_response_unit.json	urban_hru.json
Surfaces	one, infiltrating	pervious + impervious
Internal routing reaches	none	one per surface
Soil parameters	one set	one set per layer
Groundwater area	A	A
Recharge conductivity	constant <code>recharge_K</code>	$K_{sat,3} k_r(S_{e,3})$
Groundwater cell	Groundwater cell, user S_s	Unconfined Groundwater cell
Groundwater template	groundwater.json	unconfined_groundwater.json
Initial groundwater state	moisture content	water-table elevation
Evapotranspiration	Soil_1	Soil_1; optional second source on Soil_2–Soil_3
Calibration multipliers	—	impervious_multiplier, K_sat_scale, K_sat_scale_surface
Icon		

Table 1: The two HRU composites. A is the plan area, ϕ the groundwater porosity and b the groundwater thickness.

Use the **HRU** for land without significant impervious cover — forest, fields, parks, hillslopes — and for cascades of units where the overland flow of one unit runs onto the next (Section 3). Use the **Urban HRU** wherever part of the land is impervious and drains directly to a stream or storm drain: it partitions the area, routes each surface separately, and its subsurface (Section 4) stays physically sensible when the water table lies well below the soil column.

2 Elements shared by both units

Both units build their subsurface from the same members: three **Soil** layers and a **Groundwater cell**, connected by infiltration, two percolation links and a recharge link. The geometry and the soil-water relations below apply to both; the units differ in the plan areas they give these members, the recharge link and the groundwater storage (Sections 3 and 4).

2.1 Vertical geometry

The user gives three levels; everything else is derived:

Member	Thickness	Bottom elevation
Catchment, Impervious_Catchment, reaches	—	z_s (<code>surface_elevation</code>)
Soil_1	$D/7$	$z_s - D/7$
Soil_2	$2D/7$	$z_s - 3D/7$
Soil_3	$4D/7$	$z_s - D$
Groundwater	b (<code>groundwater_thickness</code>)	$z_s - D - b$

D is `depth_to_groundwater`. The 1 : 2 : 4 split puts the thinnest layer at the surface, where moisture gradients are steepest. The groundwater cell sits directly below the soil column.

2.2 Soil water

Each soil layer stores water $V = \theta A_\ell d_\ell$ and has the van Genuchten–Mualem relations

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r}, \quad \psi = -\frac{1}{\alpha} (S_e^{-1/m} - 1)^{1/n}, \quad k_r = \left[1 - (1 - S_e^{1/m})^m\right]^2, \quad m = 1 - \frac{1}{n}, \quad (1)$$

with S_e bounded to $[0.001, 0.999]$, the suction $|\psi|$ capped at 10^4 m, and the total head $h = z_{\text{bottom}} + d_\ell + \psi$ (plus a small excess pressure above saturation, governed by the specific storage). The layer's conductivity is $K_{sat} = K_{sat_original} \times K_{sat_scale_factor}$.

Infiltration

(`surfacewater_to_soil_link`, `surface` → `Soil_1`): $Q = \min(A_s, A_1) K_{sat,1} (h_s - h_1)/(d_1/2) \cdot d/(d + 0.001)$, i.e. Darcy flow over half the top layer, switched off smoothly when no water is ponded.

Percolation

(`soil_to_soil_link`): $Q = \bar{A} K_{eff} (h_i - h_j)/\bar{d}$ with $K_{eff} = \min(K_{sat,i}, K_{sat,j}) k_r(\max(S_{e,i}, S_{e,j}))$, mean area $\bar{A}$ and mean thickness $\bar{d}$. Flow can be upward (capillary rise).

Evapotranspiration

is withdrawn from `Soil_1` by the source connected to the `Evapotranspiration` property. With the `Evapotranspiration_Time_Series` (`Soil`) source the rate is $\text{corr_fact} \times ET_0 \times A_1 \theta/\theta_s$. The Urban HRU can also withdraw from `Soil_2` and `Soil_3` through a second source (Section 4.9).

3 HRU

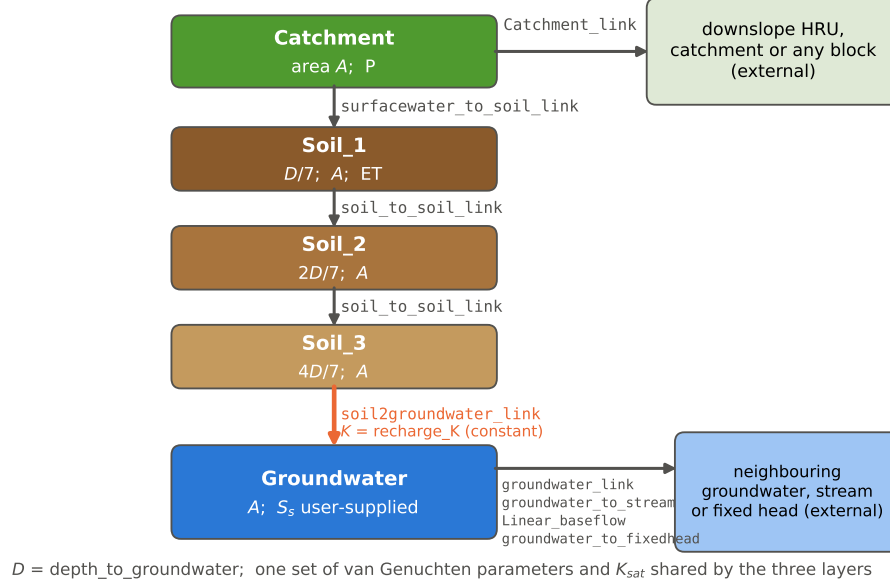


Figure 1: Hydrologic_Response_Unit: members, internal links and ports.

The HRU (`hydrologic_response_unit.json`, Fig. 1) has five members — a **Catchment** and the shared soil column and groundwater cell — and four internal links. All of them receive the whole plan area A . It has no internal reaches: the surface drains through its **Catchment_link** port directly to whatever block it is linked to, which makes cascades of units possible (the overland flow of an upslope unit runs onto the surface of the next). Its ports:

Port (link type)	From member	Typical target	Icon
Catchment_link	Catchment	a downslope HRU, a catchment, a channel or any block	
groundwater_link	Groundwater	the groundwater cell of a neighbouring unit	
groundwater_to_stream	Groundwater	a channel segment (baseflow, head-driven)	
Linear_baseflow	Groundwater	a channel segment (baseflow, linear reservoir)	
groundwater_to_fixedhead	Groundwater	a fixed_head block	

Baseflow can leave the groundwater cell through either of two connectors (use one of them for a given discharge pathway, not both in parallel):

groundwater_to_stream

Darcy flow driven by the head difference between the groundwater cell and the receiving segment, $Q = K A_b (h_g - h_c)/L$, with the connector's exchange area A_b , flow path L and

conductivity K . Discharge stops, and reverses, when the stream stage rises above the water table.

Linear_baseflow

A linear reservoir above an inactive storage (`groundwater.json`):

$$Q = k (\theta_g - \theta_{\min})_+ V_g^{\text{tot}}, \quad (2)$$

where θ_g is the moisture content and V_g^{tot} the total volume of the groundwater cell, k the `recession_rate` (1/d) and $\theta_{\min}$ the `min_moisture_content` below which no baseflow leaves (an absolute moisture content, not larger than the porosity). With no recharge the active storage recedes exponentially with time constant $1/k$. The discharge does not depend on the stream stage or on `groundwater_K`; k is set, or calibrated, directly from observed recession rates. An example is in `Examples/Linear_Baseflow/Linear_Baseflow.ohq`.

Recharge uses the stock `soil2groundwater_link` with the constant conductivity `recharge_K`,

$$Q_r = A K_{rech} \frac{h_3 - h_g}{L}, \quad L = \frac{1}{2} (d_3 + b), \quad (3)$$

and the groundwater member is the stock `Groundwater cell`, whose head is $h_g = z_g + b + (\theta_g - \phi)/S_s$ with user-supplied porosity ϕ and specific storage S_s , and whose initial state is a moisture content $\theta_{g,0}$. With the defaults ($S_s = 0.01 \text{ m}^{-1}$) the cell responds as a *confined* aquifer: a change of 0.01 in moisture content moves the head by 1 m; $S_s = \phi/b$ gives unconfined behaviour, and $\theta_{g,0} = \phi + (h_0 - z_g - b) S_s$ starts the water table at h_0 . A constant recharge conductivity is adequate when the water table lies at or just below the soil column; when it lies far below, see Section 4.6. One set of soil parameters applies to all three layers. A two-unit hillslope using the HRU is in `Examples/Hydrologic_Response_Unit/HRU_hillslope.ohq`.

4 Urban HRU

4.1 Structure

Figure 2 shows the Urban HRU (`urban_hru.json`). It has eight members:

- `Catchment` — the pervious (infiltrating) surface;
- `Impervious_Catchment` — the impervious surface, with no connection to any soil;
- `Reach`, `Impervious_Reach` — two `Trapezoidal Channel Segments` that collect the runoff of each surface: the swales, gutters, small streams and storm pipes inside the unit;
- `Soil_1`, `Soil_2`, `Soil_3` — three van Genuchten soil layers under the pervious surface;
- `Groundwater` — an `Unconfined Groundwater cell` (Section 4.7).

and six internal links: a `Catchment_link` from each surface to its reach, infiltration (`surfacewater_to_soil_link`) from the pervious surface into `Soil_1`, percolation (`soil_to_soil_link`) from `Soil_1` to `Soil_2` and from `Soil_2` to `Soil_3`, and recharge from `Soil_3` into the groundwater cell.

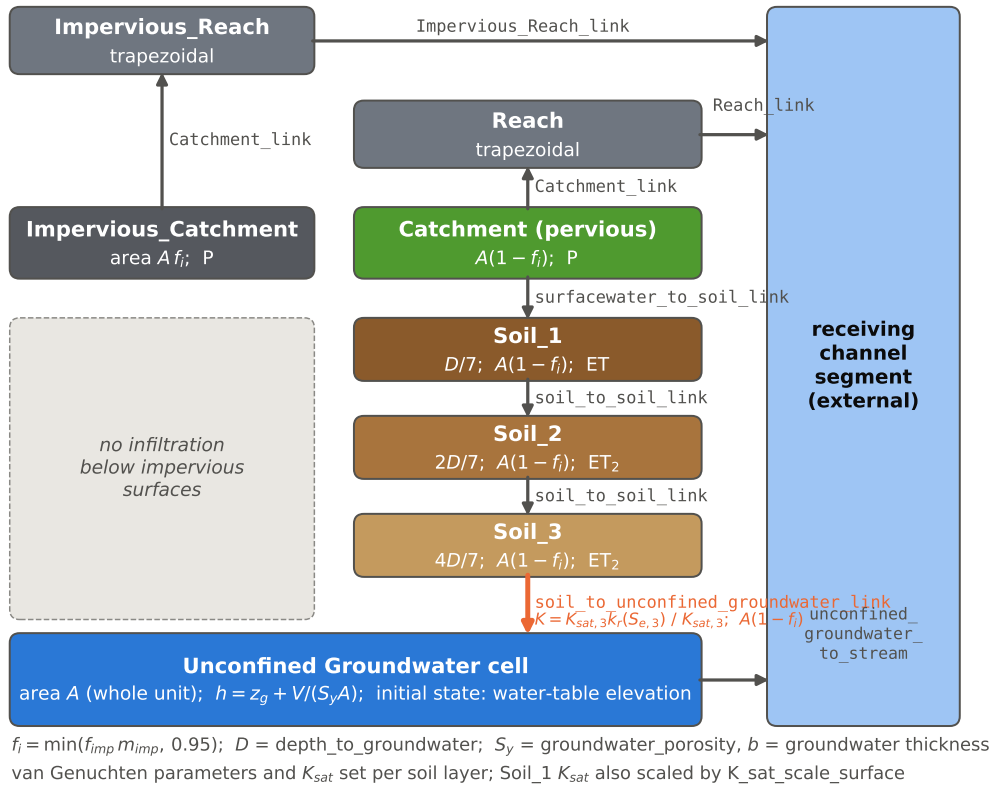


Figure 2: Urban_HRU: members (boxes, with thickness and plan area), internal links (arrows) and ports to the receiving channel segment. The recharge connector is highlighted.

4.2 Ports

A composite exposes the members that may be connected from outside as *ports*. The Urban HRU declares five, all from `unconfined_groundwater.json` or `urban_hru.json`; its baseflow leaves through `unconfined_groundwater_to_stream` (the `Linear_baseflow` connector of the HRU, Section 3, is not a port of the Urban HRU):

Port (link type)	From member	Typical target	Icon
Reach_link	Reach	the receiving channel segment	
Impervious_Reach_link	Impervious_Reach	the receiving channel segment	
unconfined_groundwater_to_stream	Groundwater	a channel segment (baseflow)	
UC_groundwater_link	Groundwater	the groundwater cell of a neighbouring Urban HRU (lateral flow)	
unconfined_groundwater_to_fixedhead	Groundwater	a fixed_head block	

`Reach_link` and `Impervious_Reach_link` are defined in `urban_hru.json`. Their equations are those of the channel-to-channel connector (`Trapezoidal_Channel_link`); they exist as separate types only so that the two outlets of the composite can be told apart when a link is drawn from it, and they have their own icons so that HRU outlets are distinguishable from links along

the stream network. When a link is drawn from the unit in the interface, the port whose type matches the chosen link is used; in a script, `create link;type=Reach_link,from=SC01,to=R01` connects the Reach member of SC01.

4.3 Surface runoff and routing

Rain falls on both surfaces. Water on a surface drains to its reach with the kinematic sheet-flow relation of `Catchment_link`,

$$Q_s = 86400 \frac{W}{n_s} S^{1/2} (d - d_s)_+^{5/3}, \quad (4)$$

with overland width W (`catchment_width`), slope S (`catchment_slope`), Manning coefficient n_s , ponded depth d and depression storage d_s ; $(x)_+ = \max(x, 0)$. Both surfaces share W and S and have their own n_s and d_s . The reaches drain to the receiving segment through the outlet ports with Manning's equation driven by the head difference,

$$Q_r = 86400 \frac{1}{\bar{n}} A_u R^{2/3} \left(\frac{h_r - h_c}{\bar{L}} \right)^{1/2}, \quad \bar{L} = \frac{1}{2} (L_r + L_c), \quad \bar{n} = \frac{1}{2} (n_r + n_c), \quad (5)$$

where A_u is the flow area of the end with the higher head, R the hydraulic radius from the mean perimeter, and subscripts r and c refer to the internal reach and the receiving channel. The reach bottoms are placed at the surface elevation.

4.4 The impervious fraction

The impervious surface drains directly to the internal impervious reach and from there to the channel, with no chance to infiltrate. `impervious_fraction` must therefore be the *directly connected* impervious area (DCIA) — roofs, roads and parking areas whose runoff reaches a storm drain or the stream without crossing pervious ground — and not the total impervious area (TIA) mapped from land cover. The composites take it as a plain number; how it is estimated is up to the modeller. Common choices:

- **Empirical TIA–DCIA relations.** Sutherland (2000) gives DCIA as a function of TIA (both in percent, for $\text{TIA} \geq 1\%$) for five degrees of connectivity:

Drainage	DCIA
Totally connected (curb, gutter and storm sewer; roofs tied in)	TIA
Highly connected	$0.4 \text{ TIA}^{1.2}$
Average (mostly storm-sewered, some roofs to lawns)	$0.1 \text{ TIA}^{1.5}$
Somewhat disconnected	$0.04 \text{ TIA}^{1.7}$
Highly disconnected (swales, roofs to lawns)	0.01 TIA^2

- **Storm-drain inventories.** Where inlet locations are available, impervious surfaces within a set distance of an inlet (or of a guttered street that leads to one) can be counted as connected. Public inventories miss private systems, so this tends to be a lower bound.
- **Calibration.** The estimate is scaled by `impervious_multiplier` (Section 4.11); keeping each unit's estimate and calibrating one multiplier preserves the spatial pattern and adjusts the level.

Impervious surfaces that are *not* directly connected remain part of the pervious surface (Section 4.12).

4.5 Areas

With $f_i = \min(f_{imp} m_{imp}, 0.95)$, where f_{imp} is `impervious_fraction` and m_{imp} the `impervious_multiplier`:

Member	Plan area
Impervious_Catchment	$A f_i$
Catchment, Soil_1–Soil_3, recharge interface	$A (1 - f_i)$
Groundwater	A

The aquifer continues beneath roads and roofs, so the groundwater cell receives the whole area; it is recharged only through the pervious soil column. Were it given only the pervious area, the water table would rise $1/(1 - f_i)$ times too fast per unit of recharge — twice as fast at $f_i = 0.5$ — and any change of f_i during calibration would also change the aquifer.

4.6 Recharge through the unsaturated conductivity

`soil_to_unconfined_groundwater_link` (`unconfined_groundwater.json`) connects `Soil_3` (the source, which must be a `Soil` block) to the groundwater cell:

$$Q_r = A(1 - f_i) K_{sat,3} \frac{k_r(S_{e,3}) (h_3 - \Delta z - h_g)_+ - (h_g - h_3 + \Delta z)_+}{L}, \quad L = \frac{1}{2} (d_3 + b - \Delta z), \quad (6)$$

with d_3 the thickness of `Soil_3` and b the `aquifer_thickness` of the cell (the unit's `groundwater_thickness`), so that L runs from the centre of the soil layer to the centre of the aquifer cell, and Δz the link's `elevation_offset` (the unit's `recharge_elevation_offset`, default 0). When the soil head is higher (drainage) the conductivity is the unsaturated conductivity of the bottom layer, with the same Mualem k_r as the percolation links; when the groundwater head is higher (upward flow) the saturated conductivity is used.

The reason is the depth of urban and upland water tables. When the water table lies several metres or more below the soil column, $h_3 - h_g$ is dominated by elevation and hardly depends on how wet the soil is. With a constant conductivity the flux is then almost independent of soil moisture: in a watershed application with the water table 6–47 m below the soil and $K_{rech} = 0.1$ m/d, the constant- K link gave a potential recharge of about 150 mm/d at $S_e = 0.5$, and at $S_e = 0.2$ the suction of the bottom layer reversed it into up to 380 mm/d of upward flow from the deep water table. Calibrating K_{rech} can only trade one error for the other. With (6) the flux falls with k_r as the bottom layer drains, and no separate recharge conductivity is needed.

Elevation offset. The soil column of a unit sits at the unit's mean surface elevation, while the single groundwater cell has its water table near the stream: in incised terrain the two can be tens of metres apart, so that recharge falls across a large gap under a nearly constant gradient and the water table can never reach the root zone. Δz lets the recharge link evaluate `Soil_3` as if it lay Δz lower, a chosen gap G above the water table, $\Delta z = (z_{3,bottom} - h_{g,0}) - G$ (at least 0); the path length shrinks by the same amount, exactly as if the column had been moved. Nothing else moves: the soil layers, the catchments, the internal reaches and the groundwater cell keep their elevations, so surface routing is unaffected. With a small G the link reverses when `Soil_3` dries (suction larger than $G + d_3$), and evapotranspiration from the deeper layers can then draw on the groundwater.

4.7 Unconfined groundwater

The groundwater member is the stock `Unconfined Groundwater` cell of `unconfined_groundwater.json`, a cell with a free water table. Its storage V_g and piezometric head are related through the specific

yield S_y (the unit's `groundwater_porosity`, which sets both `specific_yield` and `porosity` of the cell):

$$h_g = z_g + \frac{V_g}{S_y A}, \quad V_{g,0} = S_y (h_0 - z_g) A, \quad (7)$$

where z_g is the cell's `base_elevation`, so the water table rises by $1/S_y$ per unit depth of stored water. The initial state is the initial piezometric head h_0 (the unit's `initial_water_table_elevation`, same datum as `surface_elevation`) and the initial storage is computed from it; the initial water table therefore does not move when S_y changes, for example during calibration. (Specifying an initial moisture content instead would tie the initial head to S_y : a calibrated value of 0.107 instead of 0.15 would raise it by 40% of the saturated thickness.) The cell reports its saturated thickness as `depth` = $h_g - z_g$. For the Urban HRU the template gained one property, `aquifer_thickness` (the unit's `groundwater_thickness`): the thickness of the cell from its base to the bottom of the soil column. It sets the recharge path length of (6); it does not cap the storage.

Neighbouring units exchange groundwater through the stock `UC_groundwater_link`, a Darcy connector whose transmissive thickness follows the water tables:

$$Q = \frac{1}{2} (K_s + K_e) \frac{1}{2} (b_s + b_e) W \frac{h_s - h_e}{L}, \quad (8)$$

where K is the cells' `hydraulic_conductivity` (the unit's `groundwater_K`), b their saturated thickness (`depth`), W the interface width (`width`, the contact length of the shared boundary) and L the distance between the centroids of the two units (`length`). For example, `create link;type=UC_groundwater_link,from=SC01,to=SC02,width=3616,length=1352`. The flow is multiplied by a drainage cutoff on the cell it leaves (the upgradient cell, u):

$$f_c = \min\left(\frac{(b_u - b_{r,u})_+}{\delta}, 1\right), \quad (9)$$

with b_r the cell's `residual_thickness` (the unit's `groundwater_residual_thickness`, default 0) and δ the link's `cutoff_depth` (default 0.05 m). Lateral outflow therefore tapers to zero as the cell drains to its residual thickness, and a cell cannot be emptied (or driven negative) by its neighbours. Setting b_r to the depth of the cell base below the receiving stream bed keeps lateral flow from drawing the water table below the stream: with one cell per unit there is no groundwater mound under the divide between two units, so the head difference across it is the difference of their stream beds, and without the cutoff a high tributary unit can be drained into a lower neighbour until its stream turns losing. Exchange with the stream itself is not cut off.

Baseflow uses `unconfined_groundwater_to_stream`, the unconfined counterpart of `groundwater_to_stream`:

$$Q_b = K_b A_b \frac{h_g - h_c}{L_b}, \quad (10)$$

with h_c the head of the channel segment, A_b the bank (seepage-face) area (`area`), L_b the flow path to the stream (`length`) and K_b the link's `hydraulic_conductivity`. Water flows back into the aquifer when the stream is higher.

4.8 Layer-specific soil

`K_sat_i`, `alpha_vG_i`, `n_vG_i`, `theta_sat_i` and `theta_res_i` ($i = 1, 2, 3$) are set for each layer, for example from soil-survey horizons averaged over the three depth intervals. The `K_sat_i` go to each layer's `K_sat_original`, so that the layer's `K_sat_scale_factor` remains free for the multiplier below.

4.9 Evapotranspiration from the deeper soil

Besides Evapotranspiration on Soil_1, the Urban HRU has an optional source property Evapotranspiration_2, applied to Soil_2 and Soil_3 (the root zone below the top layer). Each layer's withdrawal is scaled by its own θ/θ_s . Connecting a separate source — typically the same reference-ET series with its own `corr_fact` — lets uptake from the deeper layers be scaled independently of the top layer, which gives summer evapotranspiration a way to intercept water before it drains to the groundwater cell. Left empty, the deeper layers do not transpire.

4.10 Evapotranspiration from groundwater

The Urban HRU also has an optional source property Evapotranspiration_GW, applied to the Evapotranspiration source slot of the groundwater cell (added to Unconfined Groundwater cell; its storage balance now reads recharge, Precipitation and Evapotranspiration). It is meant for a source of the new type Evapotranspiration_Time_Series (Groundwater) (`unconfined_groundwater.json`), which withdraws

$$Q_{ET,g} = c_3 E_0(t) A \frac{b_+}{b_+ + 0.01}, \quad (11)$$

with E_0 the ET series (`ET_timeseries`, m/d), c_3 the source's `corr_fact`, A the cell area and b the saturated thickness (`depth`, m). The factor is close to one for any real saturated thickness and only keeps the cell from being driven below its base. It represents riparian and deep-rooted uptake from the water table, which the soil layers above cannot supply when they sit far above it. The withdrawal applies over the whole cell area and does not depend on the depth of the water table below the surface, so c_3 should be small: a reference ET of 3.5 mm/d with $c_3 = 0.1$ removes 0.35 mm/d, comparable to or larger than summer baseflow in small Piedmont catchments. Left empty (the default), the groundwater cell does not evapotranspire and existing models are unchanged.

4.11 Calibration multipliers

All three default to 1 and leave the unit unchanged at that value.

`impervious_multiplier` (m_{imp})

scales the impervious fraction: $f_i = \min(f_{imp} m_{imp}, 0.95)$. It is meant for calibrating the *directly connected* impervious area watershed-wide when f_{imp} comes from an empirical estimate. $m_{imp} > 1$ means that more of the land drains straight to the channel than the estimate says (for example, more roofs and driveways tied to the storm drains); $m_{imp} < 1$ means more of the impervious surface is disconnected. Area moves between the impervious catchment and the pervious catchment with its soil column; the total stays A and the groundwater cell is unaffected. Infiltration, soil storage, Soil_1 evapotranspiration and recharge scale with the pervious area; the internal reaches and the `unconfined_groundwater_to_stream` link do not change. f_{imp} itself (for example a Sutherland estimate, Section 4.4) is fixed when the model is built; only m_{imp} is calibrated.

`K_sat_scale`

multiplies K_{sat} of all three layers, and therefore also the recharge conductivity of (6).

`K_sat_scale_surface`

multiplies K_{sat} of Soil_1 only: its `K_sat_scale_factor` is `K_sat_scale` $\times$ `K_sat_scale_surface`. It controls infiltration at the surface — reduced, for example, by the compaction of urban lawns and graded fill, which soil surveys of undisturbed profiles do not describe — without changing percolation and recharge through the deeper layers. Note that the infiltration link

drives flow with the suction gradient over half the top layer, so while `Soil_1` is unsaturated its infiltration capacity is several times $K_{sat,1}$; infiltration-excess runoff from the pervious surface appears only when rain intensity exceeds that capacity.

Bind a model parameter to them at the composite level,

```
create parameter; type=Parameter, name=p_imp, low=0.3, high=3, value=1, prior_distribution=log-normal
setasparameter; object=SC01, quantity=impervious_multiplier, parametername=p_imp
```

and the value reaches the members through the composite's mappings. When a parameter changes, the member areas are re-evaluated and the initial soil storages $\theta_0 A_\ell d_\ell$ recomputed, so the initial moisture content stays θ_0 ; the code generator (`ohq_generate`) compiles the same mappings and initial storages as expressions of the parameter vector.

4.12 Disconnected impervious area

Only the *connected* impervious fraction is an impervious surface. Impervious surfaces that drain onto lawns — roofs with leaders to splash blocks, driveways sloping to grass — belong to the pervious catchment $A(1 - f_i)$ and are treated as soil: rain on them can infiltrate, and the `Soil_1` evapotranspiration and the recharge interface act over them. Neither the concentration of their runoff onto the adjacent lawn nor the absence of transpiration from roofs is represented. If that matters, a third `Catchment` for the disconnected area, joined to `Catchment` by a `Catchment_link`, would represent both; it is not part of the composite.

5 Riparian Urban HRU

`Riparian_Urban_HRU` (`riparian_urban_hru.json`) is the Urban HRU with its groundwater split into two unconfined cells. A single cell per unit has one head and one flow path to the stream, so its discharge changes on one time scale (in a watershed application, about 100 days); observed baseflow, in contrast, keeps falling for one to three weeks after each event (about 10% per day at Rock Creek) before settling into the slow seasonal decline. The near-stream cell supplies the missing intermediate store.

- **Groundwater** — the regional cell, area $A(1 - f_r)$, recharged from `Soil_3` over $A(1 - f_i)(1 - f_r)$. It drains into the near-stream aquifer through an internal `UC_groundwater_link` (`hillslope_to_near_stream`; width `hillslope_link_width`, normally twice the stream length in the unit; length `hillslope_link_length`, normally the hillslope length $1/(2D_d)$), and it keeps the `UC_groundwater_link` port to neighbouring units and the fixed-head port. Optionally it also discharges directly to the stream through the `regional_groundwater_to_stream` port, a separate link type with the equation of `unconfined_groundwater_to_stream` (a composite exposes one member per link type), normally over the hillslope path; leave it unconnected for the near-stream aquifer to be the only path to the stream.
- **Near_Stream_Aquifer** — area Af_r (`near_stream_fraction`), recharged from `Soil_3` over $A(1 - f_i)f_r$ (`recharge_near_stream`), with its own specific yield (`near_stream_specific_yield`) and conductivity (`near_stream_K`). It connects to the stream through the `unconfined_groundwater_to_stream` port. With a short path (the half-width of the riparian strip) the exchange is fast and runs both ways: the stream fills it during events (bank storage) and it drains back over days to weeks.

Both cells share the base elevation, thickness, residual thickness and initial water table of the Urban HRU, and both recharge links use `recharge_elevation_offset`. There is no evapotranspiration from either cell. Everything else (surfaces, reaches, soils, ports for the surface outlets) is as in the Urban HRU. With $\tau_r = S_{y,r} A f_r L / (K_r A_b)$ for the near-stream cell and the

regional cell's drainage through the hillslope link, the unit has a fast and a slow baseflow store in series.

6 Using the units

6.1 Templates

The HRU lists `main_components.json`, `unsaturated_soil.json` and `groundwater.json` in its `requires`; the Urban HRU lists `main_components.json`, `unsaturated_soil.json`, `unconfined_groundwater.json` and `open_channel.json` (not `groundwater.json`), so selecting either alone in the interface is enough. The evapotranspiration source is not a member; load `soil_evapotranspiration_models.json` for the `Evapotranspiration_Time_Series (Soil)` source. In a script, add each template explicitly.

6.2 A worked example

`Examples/Urban_HRU/Urban_HRU_example.ohq` is one 50 ha unit with 25 % impervious area, draining to an 800 m trapezoidal stream that ends at a fixed head, forced by 30 days of hourly rain and daily reference ET:

```
loadtemplate; filename = main_components.json
addtemplate; filename = unsaturated_soil.json
addtemplate; filename = unconfined_groundwater.json
addtemplate; filename = open_channel.json
addtemplate; filename = soil_evapotranspiration_models.json
addtemplate; filename = urban_hru.json
create source;type=Precipitation,name=Rain,timeseries=precipitation.csv
create source;type=Evapotranspiration_Time_Series (Soil),name=ET,corr_fact=0.8,ET_timeseries=
    evapotranspiration.csv
create composite;type=Urban_HRU,name=Neighbourhood,area=500000,impervious_fraction=0.25,
    surface_elevation=100,depth_to_groundwater=3.5,groundwater_thickness=12,
    Precipitation=Rain,Evapotranspiration=ET,catchment_slope=0.03,catchment_width=2500,
    K_sat_1=0.6,K_sat_2=0.4,K_sat_3=0.8,alpha_vG_1=1.5,...,
    groundwater_porosity=0.15,initial_water_table_elevation=88.5,
    reach_length=400,impervious_reach_length=400,...
create block;type=Trapezoidal Channel Segment,name=Stream,base_width=3,side_slope=2,
    ManningCoeff=0.035,bottom_elevation=88,length=800,depth=0.1
create block;type=fixed_head,name=Outlet,head=87
create link;type=Reach_link,from=Neighbourhood,to=Stream,name=Pervious_outlet
create link;type=Impervious_Reach_link,from=Neighbourhood,to=Stream,name=Impervious_outlet
create link;type=unconfined_groundwater_to_stream,from=Neighbourhood,to=Stream,name=Baseflow,
    area=8000,length=150,hydraulic_conductivity=5
create link;type=channel2fixed,from=Stream,to=Outlet,name=Stream_outlet
```

(Line breaks inside `create` commands are for display only.) Members are addressed as `Neighbourhood__Soil_3`, `Neighbourhood__recharge` and so on in outputs and scripts.

Figure 3 and Table 2 show the run. All impervious runoff reaches the stream within hours of each storm, while the pervious surface produces almost none because the rain rate rarely exceeds the infiltration capacity. About three quarters of the rain infiltrates; recharge starts at 8 mm/d as the initially wet bottom layer drains, falls with its unsaturated conductivity and settles near the rate at which water percolates through the column, and the water table rises gradually and feeds baseflow between storms.

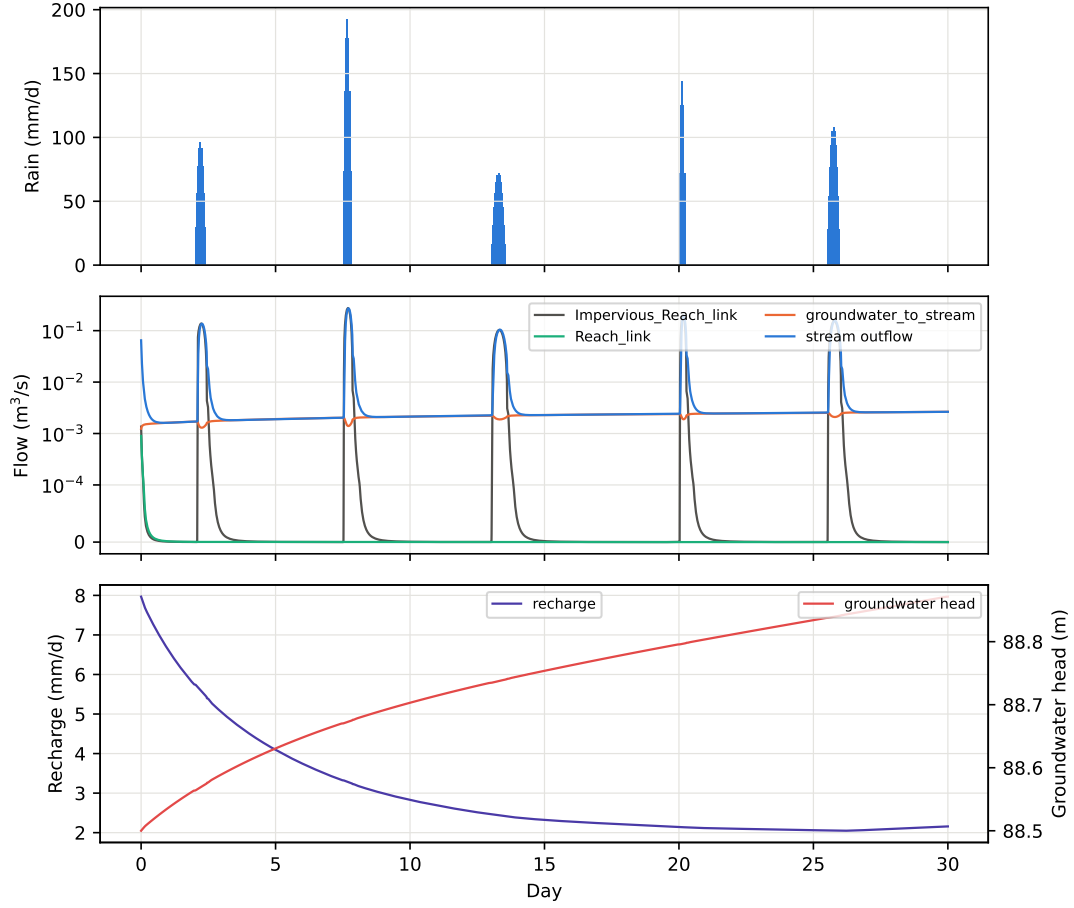


Figure 3: The worked example: rain, outflows of the three ports and of the stream, recharge and groundwater head.

Volume over 30 days	m ³
Rain	74,336
Impervious outlet (<code>Impervious_Reach_link</code>)	18,370
Pervious outlet (<code>Reach_link</code>)	4
Infiltration	55,749
Recharge	33,526
Baseflow (<code>groundwater_to_stream</code>)	5,717
Stream outflow	24,179
Groundwater head rise (m)	0.37

Table 2: Water balance of the worked example.

7 Dual drainage: units served by storm sewers

In developed catchments the runoff of a unit often has no open channel to flow in: streets and swales collect it, inlets take it into a storm-sewer trunk, and the trunk discharges into the nearest stream. Only when the sewer is full does water stay on the street and flow overland. The template `dual_drainage.json` represents this *dual drainage* with existing blocks and two new links (Fig. 4):

- a *surface segment* (any open-channel block, normally a `Compound Channel Segment` shaped as the street or swale) receives the unit's `Reach_link` and `Impervious_Reach_link`;

- a `Sewerchannelsegment` (`Sewer_system.json`) is the trunk sewer;
- `Swale_to_pipe_inlet` moves water from the surface segment into the sewer, and `Pipe_outfall_link` discharges the sewer into the receiving open channel;
- the surface segment also keeps its ordinary channel link (e.g. `Compound_Channel_link`) to the receiving channel, which carries the overflow when the inlets or the sewer cannot take the flow.

The template lists `main_components.json`, `open_channel.json`, `compound_channel.json` and `Sewer_system.json` in its `requires`. Nothing in the existing templates changes.

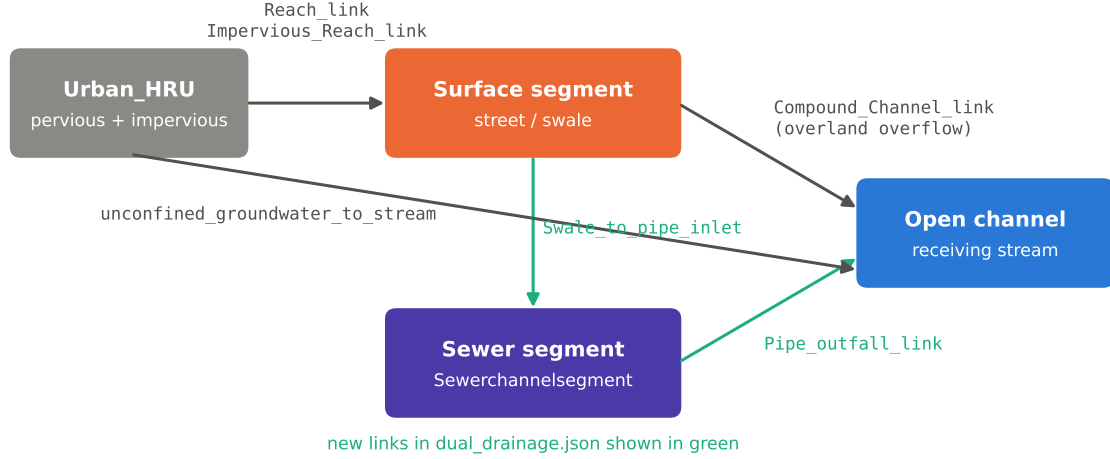


Figure 4: Dual drainage of an Urban HRU: the unit drains to a surface segment, inlets take the water into a sewer segment, the sewer discharges into the receiving channel, and flow the sewer cannot take continues overland.

7.1 Inlets

All inlets that drain the surface segment are lumped into one grate inlet with the two regimes of FHWA HEC-22: weir flow while the water over the inlets is shallow and orifice flow once they are submerged, whichever is smaller,

$$Q_w = C_w P |\Delta h|^{3/2} \frac{|\Delta h|}{|\Delta h| + h_s}, \quad Q_o = C_d A \sqrt{2g |\Delta h|}, \quad Q = \text{sign}(\Delta h) \min(Q_w, Q_o), \quad (12)$$

$$\Delta h = \max(H_{\text{surf}}, z_{\text{in}}) - \max(H_{\text{sewer}}, z_{\text{in}}), \quad (13)$$

with P the total weir length (`inlet_perimeter`), A the total effective opening (`inlet_area`), $C_w = 1.66$ (SI) and $C_d = 0.67$ (HEC-22 defaults), z_{in} the inlet elevation (normally the bottom of the surface segment) and H the hydraulic heads. Water enters while the sewer head is below the surface water level. When the sewer surcharges above it the flow reverses and the sewer overflows to the street; with both heads below the inlets the flow is zero. With `one_way` = 1 (default 0) Δh is replaced by $\max(\Delta h, 0)$: the inlets only take water in, and water that rises above the street stays in the receiving block (for a catch basin with a rim and spreading, below). The default is byte-identical to the two-way link.

The factor $|\Delta h|/(|\Delta h| + h_s)$ (`smoothing_depth` h_s , default 5 mm) matters numerically, not physically (it is below 20 % for depths above $5h_s$). A pure orifice law has an infinite derivative at zero depth, and a weir law $|\Delta h|^{3/2}$ still drains a surface segment whose storage grows as depth^2 at an unbounded relative rate as it empties; both stalled the solver in the example (the 15-day test took 111s instead of 3s). With the factor the inflow falls off as $\text{depth}^{5/2}$ and the surface segment empties smoothly.

7.2 Outfall

The sewer segment's head is at its centre; the outfall link applies Manning's equation over half its length with the sewer's flowing area A_s and wetted perimeter P_s ,

$$Q = \frac{1}{n} \text{sign}(\Delta h) \sqrt{\frac{|\Delta h|}{L/2}} A_s \left(\frac{A_s}{P_s}\right)^{2/3}, \quad \Delta h = \max(H_{\text{sewer}}, z_{\text{out}}) - \max(H_{\text{chan}}, z_{\text{out}}), \quad (14)$$

with z_{out} the outfall invert (`outfall_invert`). The outfall is free while the receiving water surface is below the invert. A full sewer segment raises its head steeply with any further storage (`Sewerchannelsegment`), which drives pressurised flow through the outfall. Reverse flow (stream above the sewer) uses the sewer's current flowing area, so it cannot fill an empty sewer from the stream.

7.3 Sewer size, surcharge and groundwater

`Sewerchannelsegment` has a `diameter_multiplier` (default 1) that scales its diameter in all of its hydraulics and in the links that read it (`Sewer2Sewer_link`, `sewer2pond`, `soil2sewer`); it lets a calibration adjust uncertain pipe sizes with one parameter while each segment keeps its own nominal diameter. With the default the results are unchanged (the example's outputs are byte-identical to those before the property was added); with 1.5 the example's sewer carries the whole second storm and the overland overflow nearly vanishes.

When a sewer is full, its head rises by `surcharge_coefficient` (m per m³) for every cubic metre stored above the full pipe. The default, 10 000, keeps the earlier behaviour of a practically rigid pipe; it is very stiff for the solver, and the code-generated kernels, which give up after 20 failed attempts at a step, stop at the first surcharge. Real sewers surcharge into their manholes: with manholes of plan area A_m every s metres the coefficient is $1/(A_m L/s)$, e.g. 0.11 for the example's 800 m sewer with 1.2 m manholes every 100 m. Once the head reaches the street the inlet link returns the water to the surface. The example uses this value.

Groundwater that enters storm sewers (infiltration through joints and trenches) can be represented with the Urban HRU's `unconfined_groundwater_to_stream` link to the sewer segment: the flow is $KA(H_{gw} - H_{sewer})/\ell$ with the sewer's head, so the sewer drains the aquifer while the water table is above it and loses water to it only while it carries flow (the link is limited at zero sewer storage; tested with the water table below and above the pipe).

7.4 Catch basins with a rim and spreading

The shafts of a sewer (manholes, inlet boxes) can be represented explicitly with `Catch basin` blocks between the inlets and the sewer (street $\rightarrow$ `Swale_to_pipe_inlet` $\rightarrow$ catch basin $\rightarrow$ `Sewer_pipe` $\rightarrow$ sewer). The block has a base area A and bottom z , and two optional properties: `max_height` h_{max} (rim above the bottom) and `spread_side_slope` m . Below the rim it is a prism; above it the plan area grows as an inverted square pyramid, $A(e) = (\sqrt{A} + 2me)^2$ at height e above the rim, which stores water that would spread over the surroundings. Its depth is

$$d = \min\left(\frac{S}{A}, h_{\text{max}}\right) + \frac{3V}{X^2 + X\sqrt{A} + A}, \quad V = \max(S - Ah_{\text{max}}, 0), \quad X = (A^{3/2} + 6mV)^{1/3},$$

the closed-form inverse of $V(e) = [(\sqrt{A} + 2me)^3 - A^{3/2}]/(6m)$, written so that it is exactly S/A when $V = 0$ and reduces to the prism when $m = 0$. With the defaults ($h_{\text{max}} = 10^{12}$, $m = 0$) the block is the earlier prism: a model using catch basins gives identical results (185 of 185 series). $dd/dS = 1/A$ on both sides of the rim, so the transition is smooth. The block reports `flood_depth` (e), `flood_volume` (V) and `spread_area` ($A(e) - A$). In the example with a catch basin of 40 m², rim 2 m, $m = 30$ and a sewer `surcharge_coefficient` of 10, the second storm raises the basin 0.16 m above the rim (20.7 m³, 211 m²); the generated kernel runs it without failure and

matches the interpreter's peak storages within 0.3 %. A catch basin alone does not remove the stiffness of a rigid sewer (default `surcharge_coefficient`): keep a finite coefficient on the sewer.

7.5 Ports

Link	From	To	Icon
<code>Swale_to_pipe_inlet</code>	surface segment	<code>Sewerchannelsegment</code>	
<code>Pipe_outfall_link</code>	<code>Sewerchannelsegment</code>	open-channel segment	

Properties: `Swale_to_pipe_inlet` takes `inlet_area`, `inlet_perimeter`, `discharge_coefficient` (0.67), `weir_coefficient` (1.66), `inlet_elevation`, `smoothing_depth` (0.005 m) and `one_way` (0); `Pipe_outfall_link` takes `outfall_invert`. Both report `head_difference` and `flow`; the outfall also reports `area` and `velocity`.

7.6 Example

`Examples/Dual_Drainage/dual_drainage_example.ohq` is a 40 ha Urban HRU (30 % impervious) draining to an 800 m street segment with inlets ($A = 1.5 \text{ m}^2$, $P = 20 \text{ m}$) into an 800 m, 0.6 m sewer that discharges into a 600 m stream; the street also overflows to the stream. Two storms test both regimes (Fig. 5; manhole surcharge storage, coefficient 0.11):

- 10 mm/h for 3 h: the inlets take all the street runoff ($0.33 \text{ m}^3/\text{s}$), the sewer runs 62 % full and nothing flows overland;
- 50 mm/h for 1 h: the street receives $1.51 \text{ m}^3/\text{s}$; the sewer surcharges (outfall $0.65 \text{ m}^3/\text{s}$ at capacity, inlets briefly $0.75 \text{ m}^3/\text{s}$ while the sewer fills) and $0.58 \text{ m}^3/\text{s}$ flows overland to the stream. The sewer head rises above the inlets but the street water surface stays higher, so no water returns to the street.

The run takes 3 s for 30 days (8 s with the rigid default). A kernel generated from the example (`codegen`) reproduces the interpreter: peak storages of street, sewer and stream within 0.5 %, time-averaged storages within 1–4 % (soils and groundwater within 0.05 %); with the rigid default the kernel stops at the second storm. Volume balances of the street and the sewer computed from the written outputs close within 0.3 % for the first storm. For the second they differ by 1–2 % at the abrupt end of the rain, where flows change faster than the output interval; with a ten times smaller time step the sewer's difference fell from 94 to 8 m^3 , i.e. it is an artefact of integrating the interpolated outputs (the links move water between blocks within each step).

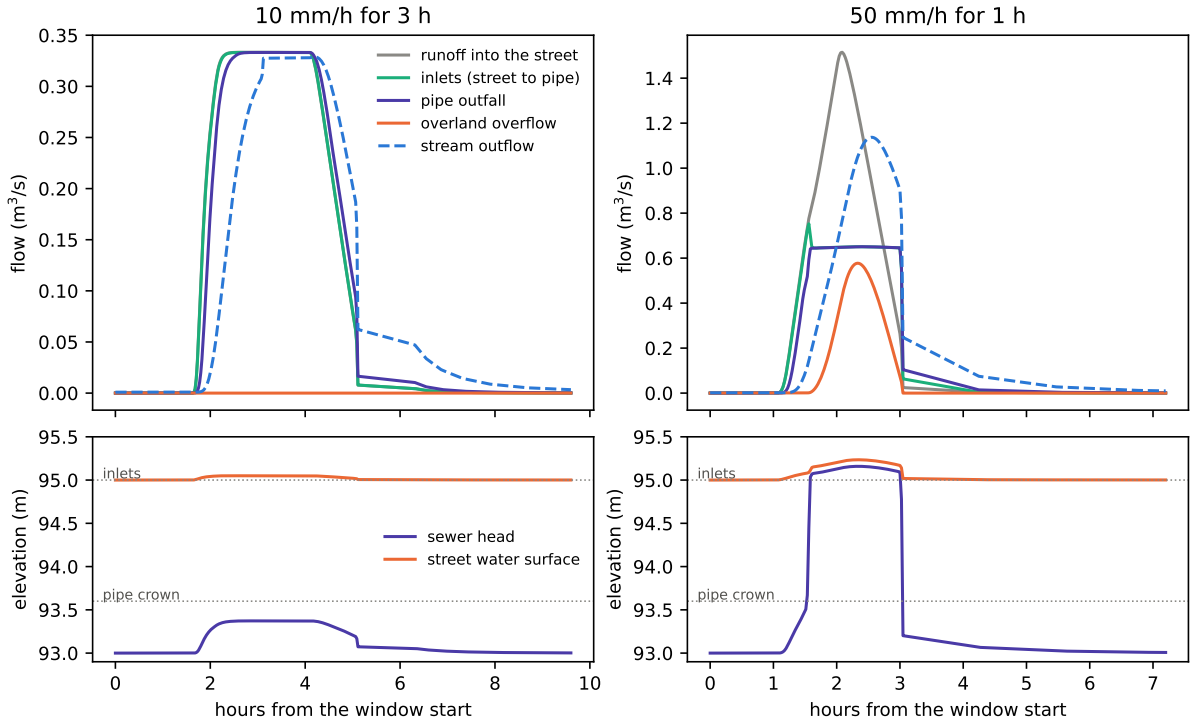


Figure 5: Dual-drainage example: flows (top) and sewer head and street water surface (bottom) for the two storms.

7.7 Limitations

- One lumped inlet per surface segment: the time water spends on the street before reaching an inlet and inlet bypass at high gutter velocities are not represented.
- One sewer segment per unit represents the trunk; the branch network's storage and travel time are lumped into it.
- The sewer cannot be filled from the receiving stream (see Outfall); backwater into the sewer is therefore underestimated during floods that rise above the outfall.

8 Property reference

Generated from the two resource files by `docs/make_hru_figures.py`. “In” gives the units that expose the property; “Applied to” lists the members that receive it; where the defaults differ between the units they are separated by a slash.

Property	Unit	Default	In	Applied to
area	m ²	1000	both	member areas
recharge_K	m/day	1	HRU	recharge
surface_elevation	m	0	both	Catchment, Groundwater, Impervious_Catchment, Impervious_Reach, Reach, Soil_1, Soil_2, Soil_3
depth_to_groundwater	m	3.5	both	Soil_1, Soil_2, Soil_3
groundwater_thickness	m	5	both	Groundwater
Precipitation			both	Catchment, Impervious_Catchment
Evapotranspiration			both	Soil_1

Property	Unit	Default	In	Applied to
catchment_slope		0.01	both	Catchment, Impervious_
catchment_width	m	30	both	Catchment, Impervious_
manning_coefficient		0.05	both	Catchment
depression_storage	m	0	both	Catchment
runoff_coefficient		1	both	Catchment, Impervious_
K_sat	m/day	1	HRU	Soil_1, Soil_2, Soil_3
alpha_vG	1/m	1	HRU	Soil_1, Soil_2, Soil_3
n_vG		1.41	HRU	Soil_1, Soil_2, Soil_3
theta_sat		0.4	HRU	Soil_1, Soil_2, Soil_3
theta_res		0.05	HRU	Soil_1, Soil_2, Soil_3
soil_specific_storage	1/m	0.01	both	Soil_1, Soil_2, Soil_3
initial_moisture_content		0.2	both	Soil_1, Soil_2, Soil_3
groundwater_K	m/day	1	both	Groundwater
groundwater_porosity		0.2/0.4	both	Groundwater
groundwater_specific_storage	1/m	0.01	HRU	Groundwater
initial_groundwater_moisture_content		0.25	HRU	Groundwater
impervious_fraction		0.2	Urban	member areas
groundwater_residual_thickness	m	0	Urban	Groundwater
recharge_elevation_offset	m	0	Urban	recharge
Evapotranspiration_2			Urban	Soil_2, Soil_3
K_sat_1	m/day	1	Urban	Soil_1
K_sat_2	m/day	1	Urban	Soil_2
K_sat_3	m/day	1	Urban	Soil_3
alpha_vG_1	1/m	1	Urban	Soil_1
alpha_vG_2	1/m	1	Urban	Soil_2
alpha_vG_3	1/m	1	Urban	Soil_3
n_vG_1		1.41	Urban	Soil_1
n_vG_2		1.41	Urban	Soil_2
n_vG_3		1.41	Urban	Soil_3
theta_sat_1		0.4	Urban	Soil_1
theta_sat_2		0.4	Urban	Soil_2
theta_sat_3		0.4	Urban	Soil_3
theta_res_1		0.05	Urban	Soil_1
theta_res_2		0.05	Urban	Soil_2
theta_res_3		0.05	Urban	Soil_3
initial_water_table_elevation	m	93	Urban	Groundwater
impervious_manning_coefficient		0.015	Urban	Impervious_Catchment
impervious_depression_storage	m	0.001	Urban	Impervious_Catchment
reach_base_width	m	1	Urban	Reach
reach_side_slope		2	Urban	Reach
reach_manning_coefficient		0.035	Urban	Reach
reach_length	m	10	Urban	Reach
initial_reach_depth	m	0.01	Urban	Reach
impervious_reach_base_width	m	1	Urban	Impervious_Reach
impervious_reach_side_slope		2	Urban	Impervious_Reach

Property	Unit	Default	In	Applied to
impervious_reach_manning_coefficient		0.02	Urban	Impervious_Reach
impervious_reach_length	m	10	Urban	Impervious_Reach
initial_impervious_reach_depth	m	0.01	Urban	Impervious_Reach
impervious_multiplier		1	Urban	member areas
K_sat_scale		1	Urban	Soil_1, Soil_2, Soil_3
K_sat_scale_surface		1	Urban	

9 Limitations and pitfalls

- **Fixed layer split.** The three soil layers are always $D/7$, $2D/7$ and $4D/7$; D controls the storage, the recharge path length and the heads together, so calibrating it moves all three at once.
- **Deep water tables in the HRU.** When the water table is far below the soil column, the HRU's constant recharge conductivity drains the column regardless of its wetness and pulls water upward when the bottom layer dries (Section 4.6); keep `recharge_K` small or use the Urban HRU, whose recharge follows the unsaturated conductivity.
- **Confined-style defaults in the HRU.** With $S_s = 0.01 \text{ m}^{-1}$ the head responds metres to small changes of storage. For an unconfined aquifer set $S_s = \phi/b$.
- **Outlet link length.** (5) averages the length of the internal reach with that of the receiving segment, so a long receiving segment lowers the outlet slope.
- **Internal reaches are generic.** Their shape and roughness are properties of the unit, not derived from its drainage network; they mostly control the timing of the runoff peak.
- **Evapotranspiration.** The HRU transpires from `Soil_1` only; the Urban HRU from `Soil_1` and, with `Evapotranspiration_2`, from `Soil_2` and `Soil_3`. Each layer's rate is scaled by its own θ/θ_s . The groundwater cell transpires only with `Evapotranspiration_GW` (Section 4.10), at a rate independent of the water-table depth.
- **Hourly rainfall and infiltration excess.** With hourly forcing, short bursts of high intensity are averaged out, so infiltration-excess runoff from pervious surfaces is underestimated; `K_sat_scale_surface` can compensate in calibration.
- **Disconnected impervious area** is treated as pervious soil (Section 4.12).
- **One unit, one soil column.** Spatial variation inside a sub-catchment is represented only through its areas; use more units for more variation.

Reference

Sutherland, R. C. (2000). Methods for estimating the effective impervious area of urban watersheds. *Watershed Protection Techniques* 2(4), 282–284.